

Medical Device Evaluation and Safety Report

Quality Assurance and Regulatory Affairs

September 12, 2026

1 Device Identification

Parameter	Details
Device Trade Name	RetinaAI-Dx Diagnostic Engine
Model / Catalog Number	v2.4.1 (Build 2026.1)
Device Classification	Class 3 / ASEAN CSDT Rule 10
Manufacturer	HealthTech Solutions Ltd.
Document Control ID	MDR-2026-001-REV02

2 Executive Summary

This report documents the verification, clinical evaluation, and regulatory compliance status of the device specified above. The device fulfills essential safety and performance requirements under applicable medical device directives and international quality standards for Thai FDA active diagnostic software registration.

3 Design Verification and Performance Testing

Bench testing was conducted under standard environmental conditions. All primary performance indicators met predefined acceptance criteria.

Test Item	Standard / Protocol	Acceptance Criteria	Result
Electrical Safety	IEC 60601-1	Leakage current $< 100 \mu\text{A}$	Pass
EMC Compatibility	IEC 60601-1-2	Radiated emissions Class B	Pass
Software Validation	IEC 62304 (Class B)	Zero critical defects	Pass
Biocompatibility	ISO 10993-1	Cytotoxicity score ≤ 1	Pass

4 Risk Analysis Summary (ISO 14971)

A comprehensive risk analysis was executed across the device lifecycle. Risk control measures were implemented for all identified hazards, reducing residual risks to acceptable levels.

- **Hazard 1: Software Processing Error** – Mitigated via redundant calculation routines and integrity checks; residual risk evaluated as broad acceptable.
- **Hazard 2: Ungradable Optical Inputs / Artifacts** – Mitigated through integrated quality gates, contrast enhancement, and automated image rejection prompts.
- **Hazard 3: Operator Interface Misinterpretation** – Mitigated by standardized GUI alerts and human-factors engineering (IEC 62366-1).

5 Regulatory and Post-Market Surveillance Declaration

The device complies with all applicable essential principles of safety and performance. Post-Market Clinical Follow-up (PMCF) activities are scheduled biannually to monitor clinical outcomes, algorithmic drift, and adverse event logs.

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Date: September 12, 2026

Approved By: Regulatory Affairs Director
Date: September 12, 2026